System Test Plan

Instructions. In this section, you must provide your system test plan with at least 5 test cases. **If you want to provide more than 5 test cases, add an appendix at the end of this document with the 6th, 7th, etc. test cases so that page numbers for all sections match the required template! Only the first 5 test cases will be graded.**

Make sure:

- You provide your sample test data
- Test IDs are uniquely identified and descriptive
- Test descriptions are fully specified with complete inputs, specific values, and preconditions
    - Be sure to provide SPECIFIC INPUTs and VALUEs so that your test cases are repeatable
- Expected results are fully specified with specific output values
- All tests cover scenarios based on the problem statement
- All tests cover unique scenarios for the system
- All strategies for system testing are demonstrated in the tests (testing equivalence classes, testing boundary values, testing exceptions/unexpected inputs)

Test Data:

- **landmarks1_test.txt**
  LANDMARK_ID,DESCRIPTION,TYPE
  L01,Park Entrance,Location
  L02,Entrance Fountain,Fountain
  L03,Waste Station 1,Pet Waste Station
  L04,Entrance Restrooms,Restroom
  L05,Overlook 1,Overlook
  L06,Rock Formation 1,Rock Formation
  L07,Overlook 2,Overlook
  L08,Overlook Restrooms,Restroom

- **trails1_test.txt**
  LANDMARK_ID,LANDMARK_ID,DISTANCE
  L01,L02,500
  L01,L03,800
  L02,L04,1200
  L03,L05,900
  L04,L06,750
  L05,L07,600
  L06,L08,300

Test ID	Description	Expected Results	Actual Results
Test #1 **testID: testUC1Valid** **Strategy: Equivalence class (load valid input files)**	**Preconditions: landmarks_test.csv and trails_test.csv exist** **Steps:** 1. **Run TrailManager** 2. **Enter landmarks1_test.txt and trails1_test.txt when prompted**	Files load successfully with no errors	Files load and system displays message asking what to do next
Test #2 **testID: testUC1Invalid** **Strategy: Exception (load non-existent input files)**	**Preconditions: invalid_file.txt doesn't exist** **Steps:** 1. **Run TrailManager** 2. **Enter invalid_file.csv for the landmarks file**	System re-prompts the user to specify a new file.	The system displays an error message and re-prompts the user for a new file

Test #3
 	
testID:testUC2FirstAid
 	
Strategy: Equivalence class (list first aid stations)
	Preconditions: UC1 completed successfully
	
	Steps:
	
	1. **Select "List first aid locations"**
	2. **Enter 2 for the requested number**
		The system displays the following message:
		
		Proposed Locations for First Aid Stations {
		
		Entrance Fountain (L02) - 2 intersecting trails
		
		Entrance Restrooms (L04) - 2 intersecting trails
		
		Overlook 1 (L05) - 2 intersecting trails
		
		Park Entrance (L01) - 2 intersecting trails
		
		Rock Formation 1 (L06) - 2 intersecting trails
		
		Waste Station 1 (L03) - 2 intersecting trails
		
		}
			The system correctly displays the potential first aid stations in the correct order

Test #4 **testID: testUC2InvalidMin** **Strategy: Boundary value (invalid minimum number)**	**Preconditions: UC1 completed successfully** **Steps:** 1. **Select "List first aid locations"** 2. **Enter 0 for the requested number**	System displays the error message "Number of intersecting trails must be greater than 0."	System correctly displays the error message

Test #5 **testID: testUC2InvalidMax** **Strategy: Exception (invalid maximum intersections)**	**Preconditions: UC1 completed successfully** **Steps:** 1. **Select "List first aid locations"** 2. **Enter 4 for the requested number**	The system displays the message "No landmarks have at least 4 intersecting trails."	System correctly displays the error message

Test #6
 	
testID: testUC3Distance
 	
Strategy: Equivalence class (Calculate distances from landmark)
	Preconditions: UC1 completed successfully
	
	1. **Select "View distance"**
	2. **Enter L02**
		The system displays the message:
		
		Landmarks Reachable from Entrance Fountain (L02) {
		500 feet to Park Entrance (L01)
		1200 feet to Entrance Restrooms (L04)
		1300 feet to Waste Station 1 (L03)
		1950 feet to Rock Formation 1 (L06)
		2200 feet to Overlook 1 (L05)
		2250 feet to Overlook Restrooms (L08)
		2800 feet to Overlook 2 (L07)
		}
			The correct message is displayed with the correct formatting

Test #7 **testID: testUC3InvalidID** **Strategy: Exception (Nonexistent ID)**	**Preconditions: UC1 completed successfully** 1. **Select “View distance”** 2. **Enter L99**	System displays the error message “The provided landmark ID (L99) is invalid for the park.”	The system correctly displays the error message